

Delivering 9x7 L2 Application and DBA Support for a Leading Nutrition and Wellness Brand

Introduction

A prominent U.S.-based nutrition company specializing in weight management products and services operates under a direct-to-consumer model, selling through online, mobile, and telephonic channels

For over 18 years, NuWare has been a strategic technology partner for this client, supporting their eCommerce platform, mobile applications, and enterprise back-office systems. As the company expanded its digital ecosystem, it required a dependable partner to provide round-the-clock application and database support to ensure consistent uptime, seamless performance, and efficient incident resolution.

In response, NuWare designed and implemented a 9x7 offshore L2 Application and DBA support model, leveraging cloud-native technologies, proactive monitoring, and SLA-driven governance. The engagement ensures continuous system availability and faster response times across the client's mission-critical platforms hosted on both on-premises infrastructure and AWS Cloud.

The Challenge

As the client scaled operations across multiple platforms and data environments, several key challenges emerged:

1. **Complex Hybrid Infrastructure**

A distributed environment spanning on-premise servers and AWS cloud required synchronized management of application, database, and integration layers.

2. **Uptime and Performance Monitoring**

Maintaining optimal performance for customer-facing eCommerce systems demanded 24x7 proactive monitoring and near-instant resolution of incidents.

3. **Critical Job Failures and Delays**

Scheduled reports, data jobs, and batch processes often failed or experienced latency, impacting business workflows and reporting accuracy.

4. **Multi-RDBMS Environment Management**

The client's applications ran on a diverse set of databases, Microsoft SQL Server, Oracle, PostgreSQL/Aurora RDS, and Snowflake, necessitating specialized DBA expertise.

5. **Coordination Across Multiple Teams**

Efficient problem resolution required cross-functional collaboration between internal teams, third-party vendors, and NuWare's offshore support center.

6. **Tight SLA Commitments**

The client demanded a **15-minute response window** for all Priority 1 alerts and 9x7 operational coverage across multiple time zones.

These challenges underscored the need for a scalable, technology-agnostic support model capable of handling both application performance and database health while ensuring end-to-end visibility and control.

NuWare's Approach

NuWare implemented a hybrid L2 Application and DBA support framework combining proactive monitoring, incident response, and performance optimization. This model was designed to ensure continuous uptime, timely escalation, and seamless collaboration across teams.

Engagement Overview

- **Year Started:** 2019 – Ongoing
- **Average Team Size:** 5 engineers
- **Support Window:** 11:00 PM – 8:00 AM EST (9x7 coverage)
- **Technologies Supported:** Apache Webserver, JBoss, IIS, OEM, OMS, CertAdmin, Business Objects, Rubrik, New Relic, PagerDuty, Jira, Confluence, FiveTran, RPI, RPDM.
- **Database Platforms:** Microsoft SQL Server, Oracle, PostgreSQL/Aurora RDS, Snowflake.
- **Infrastructure:** On-premises and AWS Cloud.

L2 Application Support

NuWare's **L2 Application Support** team provides **9x7 monitoring and operational management** of application servers, databases, and batch processes, ensuring uninterrupted business continuity.

Key Responsibilities:

- **Server & Application Alerts:** Continuous SLA-based monitoring of server health, application status, and performance warnings.
- **Health Checks:** Regular validation of application services, job completion, and system stability.

- **Batch and Report Management:** Monitoring and resolving slowness or failure in critical batch jobs and reports.
- **Application Requests:** Executing scheduled requests such as report generations, file transfers, and parameter updates.
- **DB Refresh Management:** Overseeing refresh operations across **Integration and UAT environments**, resolving any failures or delays.
- **SSIS/SSRS Deployments:** Managing deployments, troubleshooting, and performance tuning of SQL Server Integration and Reporting Services.
- **Documentation:** Maintaining updated **SOPs, runbooks, and incident logs** to ensure operational transparency and knowledge continuity.

This comprehensive monitoring framework ensured **minimal downtime** and rapid incident response across all production and non-production environments.

L2 DBA Support

NuWare's **L2 DBA Support** complements the application monitoring layer by providing business-hour database administration and optimization services. The DBA team ensures that application databases perform reliably, securely, and efficiently.

Core Responsibilities:

- **Requirement Analysis & Solutioning:** Understanding application database dependencies and executing tailored solutions.
- **Job Failure Management:** Troubleshooting and resolving database job failures and escalating unresolved issues when necessary.
- **Code Management:** Deploying and validating **code changes** across QA and UAT environments.
- **Performance Tuning:** Conducting **server-level and query-level optimizations** to enhance database response times.
- **Advanced DBA Activities:** Planning maintenance and optimization activities, including **Daylight Saving Time adjustments** and schema updates.
- **User Service Requests:** Managing security permissions, creating/modifying database objects, and supporting development teams with schema enhancements.

The DBA team's proactive monitoring and tuning efforts significantly reduced query response times and improved overall system throughput.

Supported Infrastructure and Applications

NuWare supports both **Backoffice** and **eCommerce** systems, managing applications, databases, and integration tools across **on-premises and AWS** environments.

Environment	Type	Services / Applications	Prod	Non-Prod	Total
On-Premise	Application / Services	OMS, MSMQ, IIS, BO, SharePoint, SAP, MAS500, NSA	25	41	66
On-Premise	Database	MSSQL, Replication, SSIS, SSAS, CDC	12	20	32
AWS Cloud	Application / Services	CDP, RPI, RPDM, FiveTran, Responsys, Five9	8	11	19
AWS Cloud	Database	MSSQL, Oracle, Aurora PostgreSQL, Snowflake	10	14	24
Third-Party / PaaS	Services	FiveTran, Five9, PagerDuty, Seeburger, Aspect, Akamai, Responsys, CyberSource	N/A	N/A	N/A

This hybrid setup enables agile deployments, fast data synchronization, and fault-tolerant scalability across cloud and on-premises systems.

Outcomes

The partnership has yielded measurable improvements in application reliability, performance, and operational agility.

1. Enhanced Uptime and Stability

- Achieved **99.8% uptime** across eCommerce and Backoffice systems.
- Reduced job failure incidents through proactive monitoring and root cause analysis.

2. Faster Incident Response

- Maintained **<15-minute SLA response time** for Priority 1 alerts.

- Improved Mean Time to Resolution (MTTR) by 25% through cross-team coordination.

3. Improved Performance and Scalability

- Optimized database queries, improving average response times by **30%**.
- Ensured seamless scaling of AWS environments for traffic surges during seasonal campaigns.

4. Operational Efficiency and Cost Reduction

- Centralized monitoring reduced manual intervention by 40%.
- Delivered consistent **9x7 support coverage** at significantly lower operational costs.

5. Strong Governance and Documentation

- Standardized SOPs and reporting improved audit readiness and knowledge transfer.

Future Outlook

NuWare continues to evolve the support model to incorporate **next-generation automation and observability**:

- **Predictive Monitoring:** Integrating AI-based analytics to identify anomalies before they affect performance.
- **Self-Healing Systems:** Implementing auto-restart and self-recovery mechanisms for critical jobs.
- **Enhanced Cloud Integration:** Extending automation for Aurora RDS and Snowflake workloads.
- **DevOps Alignment:** Embedding L2 support processes within CI/CD pipelines for faster deployment cycles.

These enhancements will transform the client's operations into a self-optimizing support ecosystem, capable of sustaining growth and innovation.